

5 Contrapositive Proof

A. Use the method of contrapositive proof to prove the following statements.

2. Suppose $n \in \mathbb{Z}$. If n^2 is odd, then n is odd.

Proof. We prove the contrapositive: if n is not odd, then n^2 is not odd.
Suppose that n is not odd. Thus n is even, so $n = 2a$ for some integer a . Consequently $n^2 = (2a)^2 = 4a^2 = 2(2a^2)$. Therefore $n^2 = 2b$, where the integer $b = 2a^2$. Consequently, n^2 is even by definition of an even number. Therefore n^2 is not odd. $\square$

4. Suppose $a, b, c \in \mathbb{Z}$. If a does not divide bc , then a does not divide b .

Proof. We prove the contrapositive: if $a \mid b$, then $a \mid bc$.
Suppose $a \mid b$. Then $b = ak$ for some $k \in \mathbb{Z}$. Multiplying both sides by c gives $bc = a(kc)$. Since $kc \in \mathbb{Z}$, it follows that $a \mid bc$. $\square$

6. Suppose $x \in \mathbb{R}$. If $x^3 - x > 0$ then $x > -1$.

Proof. We prove the contrapositive: if $x \leq -1$ then $x^3 - x \leq 0$.
Suppose $x \leq -1$. Then $x < 0$, $x - 1 < 0$, and $x + 1 \leq 0$. Notice that $x^3 - x = x(x - 1)(x + 1)$, a product of two negative factors and one nonpositive factor. Since the product of two negative numbers and a nonpositive number is nonpositive, we have $x(x - 1)(x + 1) \leq 0$, that is, $x^3 - x \leq 0$. $\square$

8. Suppose $x \in \mathbb{R}$. If $x^5 - 4x^4 + 3x^3 - x^2 + 3x - 4 \geq 0$, then $x \geq 0$.

Proof. We prove the contrapositive: If $x < 0$ then $x^5 - 4x^4 + 3x^3 - x^2 + 3x - 4 < 0$. Suppose $x < 0$. Then $x = -k$ for some $k \in \mathbb{R}$ and $k > 0$. Substituting into the polynomial gives $(-k)^5 - 4(-k)^4 + 3(-k)^3 - (-k)^2 + 3(-k) - 4 = -k^5 - 4k^4 - 3k^3 - k^2 - 3k - 4 = -(k^5 + 4k^4 + 3k^3 + k^2 + 3k + 4)$. Since $k > 0$ then $k^5 + 4k^4 + 3k^3 + k^2 + 3k + 4 > 0$. Therefore $-(k^5 + 4k^4 + 3k^3 + k^2 + 3k + 4) < 0$, that is, $x^5 - 4x^4 + 3x^3 - x^2 + 3x - 4 < 0$. $\square$

10. Suppose $x, y, z \in \mathbb{Z}$ and $x \neq 0$. If $x \nmid yz$ then $x \nmid y$ and $x \nmid z$.

Proof. We prove the contrapositive: if $x \mid y$ or $x \mid z$, then $x \mid yz$. Suppose it is not true that $x \nmid y$ and $x \nmid z$. Then it is not true that $x \nmid y$, or it is not true that $x \nmid z$. Therefore $x \mid y$ or $x \mid z$. We consider these possibilities separately.

Case 1. Suppose $x \mid y$. Then $y = xk$ for some $k \in \mathbb{Z}$. Multiplying both sides by z we get $yz = x(kz)$, thus $x \mid yz$.

Case 2. Suppose $x \mid z$. Then $z = xm$ for some $m \in \mathbb{Z}$. Multiplying both sides by y we get $zy = x(my)$, thus $x \mid yz$.

The two cases above both show that $x \mid yz$. Therefore it is not true that $x \nmid yz$. $\square$

12. Suppose $a \in \mathbb{Z}$. If a^2 is not divisible by 4, then a is odd.

Proof. We prove the contrapositive: if a is not odd, then a^2 is divisible by 4. Suppose a is not odd. Thus a is even, so $a = 2k$ for some $k \in \mathbb{Z}$. Then $a^2 = (2k)^2 = 4k^2$. Since $k^2 \in \mathbb{Z}$, it follows that a^2 is divisible by 4. $\square$